THE TRANSFORMATION FROM MORPHISMS TO THE HILB FUNCTOR IS REPRESENTABLE BY OPEN IMMERSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma quot-lemma-Mor-into-Hilb (canonical tag 0D1A). cited at statement line 3394. Notation as in the collected Mor-into-Hilb card.
- Lemma spaces-more-morphisms-lemma-where-isomorphism (canonical tag 05XD). cited at proof line 3406. Openness of the isomorphism locus.

Source reference. Original lemma 0D1A: sources/stacks-chow/tags.txt, source lines 16340–16340. Cited at statement line 3394: notation and the Mor-into-Hilb transformation.

Source reference. Original lemma 05XD: sources/stacks-chow/tags.txt, source lines 7617–7617. Cited at statement line 3406: the locus where k: Y -> Z_T is an isomorphism is open.

Lemma 0.2. *Assumption and notation as in Lemma 0D1A. The transformation* $Mor_B(Z, X) \longrightarrow \mathrm{Hilb}_{Z \times_B X/B}$ *is representable by open immersions.*

Proof. Let T be a scheme over B and let $Y \subset (Z \times_B X)_T$ be an element of $\mathrm{Hilb}_{Z \times_B X/B}(T)$. Then we see that Y is the graph of a morphism $Z_T \to X_T$ over T if and only if $k = \mathrm{pr}_1|_Y : Y \to Z_T$ is an isomorphism. By More on Morphisms of Spaces, Lemma 05XD there exists an open subscheme $V \subset T$ such that for any morphism of schemes $T' \to T$ we have $k_{T'} : Y_{T'} \to Z_{T'}$ is an isomorphism if and only if $T' \to T$ factors through V. This proves the lemma. □

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